

Modelling the Excess Mortality Associated with Heat Waves in Hong Kong: 2014-2023

Zhenyuan Liu¹, Chao Ren², Jingwen Liu³, Kawasaki Yurika¹, David Makram Bishai¹

¹School of Public Health, The University of Hong Kong, Hong Kong SAR, China

²Faculty of Architecture, The University of Hong Kong, Hong Kong SAR, China

³School of Public Health, Faculty of Health and Medical Sciences, The University of Adelaide, Adelaide, Australia

*Corresponding author

David Makram Bishai

Email: dbishai@hku.hk

Abstract

Introduction

Heat waves are increasingly frequent and linked to higher mortality risks in Hong Kong. However, estimates of total excess mortality associated with heat waves remain unavailable. This study quantifies excess deaths associated with heat waves in Hong Kong from 2014 to 2023.

Methods

Daily age- and sex-specific mortality rates and population data were obtained from the Hong Kong Life Tables and Census and Statistics Department. Temperature data came from the Hong Kong Observatory, and relative risks were derived from local research. A Monte Carlo simulation was used to estimate heat-attributable deaths under different heat wave definitions, calculating total excess deaths and annualized death rates per 100,000 population.

Results

Between 2014 and 2023, heat exposure resulted in an estimated 1,455 (95% CI: 1,098–1,812) to 3,238 (95% CI: 3,234–3,242) excess deaths. In 2023, annualized excess death rates ranged from 2.95 (95% CI: 2.41–3.50) to 5.09 (95% CI: 5.07–5.12) per 100,000 people. Males and individuals aged 65 or older were disproportionately affected.

Conclusion

Over the 10-year study period, 1,455 to 3,238 excess deaths in Hong Kong were attributed to extreme heat. Heat waves now rank among the top ten causes of death in Hong Kong, with mortality rates comparable to diabetes. These findings underscore the need for urgent public health interventions to mitigate the impact of extreme heat.

Keywords: Extreme Heat, Excess Deaths, Relative Risk, Hong Kong

Author Summary

Why was this study done?

- Hong Kong has experienced increasingly frequent and intense heatwaves in recent years, yet the health impacts of extreme heat have not been comprehensively quantified.
- There is a lack of localized evidence on the mortality burden attributable to extreme heat, particularly across different demographic groups, to inform public health responses in subtropical urban settings.

What did the researchers do and find?

- This study modelled the association between extreme heat and all-cause mortality in Hong Kong from 2014 to 2023, applying multiple heatwave definitions.
- The results estimate that over 1,400 excess deaths could have been preventable during the study period, with elevated vulnerability observed among males and individuals aged 65 years and older.
- Extreme heat was identified as an independent risk factor for mortality, with a health burden comparable to that of chronic diseases such as diabetes.

What do these findings mean?

- The findings underscore the urgent need for Hong Kong to develop and implement comprehensive Heat Action Plans (HAPs) to reduce excess mortality during extreme heat events.
- With climate change expected to increase the frequency and severity of heatwaves—and an aging population—HAPs must prioritize the protection of high-risk groups.
- These results provide critical evidence for public health planning, emphasizing the importance of tailored, data-driven interventions to enhance climate resilience in urban populations.

Introduction

The intensification of climate change in recent years has resulted in more frequent heatwaves.¹ According to a recent World Meteorological report, there is an 80 percent likelihood that the annual average global temperature will temporarily exceed 1.5°C above pre-industrial levels for at least one of the next five years,² making exposure to extreme heat a growing public health concern. Extreme heat disrupts the body's ability to regulate temperature, leading to serious health consequences and exacerbating chronic conditions, which increases mortality risk that were well-documented by previous studies.^{3, 4, 5} Furthermore, the adverse effects are exacerbated in cities with high population density and outmoded urban architecture, such as poorly insulated buildings and inadequate ventilation systems, where urban heat islands further intensify indoor temperatures during heat waves.^{6, 7}

There is now compelling evidence to demonstrate the general relationship between temperature and mortality, as observed in various settings worldwide, including Hong Kong. These studies use daily time series data on deaths and temperatures and estimate the probability of dying on or after a hot day in terms of relative risks,^{8, 9, 10, 11} attributable fractions,⁹ and excess mortality rate.^{12, 13} However, absolute numbers of how many people died are more effective than relative risks in conveying the total

burden of mortality during extreme heat events to the public and stakeholders.¹⁴ Some studies have reported the number of heat-attributable deaths,^{9, 15} but attributing deaths solely to heat is complex and may underestimate heat-related mortality, as heat exposure can exacerbate pre-existing medical conditions, thereby increasing deaths from causes beyond heatstroke and hyperthermia.¹⁶ Fewer studies estimate excess deaths during heatwaves—calculated as the difference between observed and expected deaths under normal conditions—an approach that mitigates ascertainment bias from death records that often fail to attribute heat as the underlying cause.^{17, 18, 3} Given the above, excess death measures are advantageous due to their objectivity, replicability, and high face validity.

To quantify the excess mortality associated with heatwaves and to effectively communicate the health implications of prolonged elevated temperatures, we utilized published estimates of relative risk (RRs) of mortality during heat events specific to Hong Kong.^{9,11} By estimating the absolute number of excess deaths and the age-standardized excess death rate per 100,000 population, we provide a critical metric for assessing the mortality burden associated with heat exposure.

Method

Definition of heat waves

Given the absence of a universally accepted definition for a heat wave, we employed four distinct definitions that capture heat wave conditions in Hong Kong, as delineated in previous local studies (**Table 1**). The first definition identifies a heat wave as a 20-day period commencing when the daily mean temperature exceeds the 99th percentile threshold of 30.60 °C. This duration accounts for the lagged effects of a single hot day, which may persist for up to 20 days, as estimated by Liu et al. (2020).⁹ Additionally, to qualify as an independent heat wave event, the initial day must be separated by at least 20 days from the preceding day that surpassed the threshold. The second definition characterizes a heat wave based on the number of consecutive days with daily maximum temperatures exceeding 33 °C. The third definition focuses on the number of consecutive days with daily minimum temperatures above 28 °C. Lastly, the fourth definition determines a heat wave by the simultaneous occurrence of hot days and hot nights, following the metrics established by Wang et al. (2019).¹¹ These diverse criteria

were selected for our analysis based on their well-established associations with increased excess mortality risks, while controlling for lagged effects and common confounding factors such as humidity, air pollutants, and seasonality.

Daily mean temperatures from 2014 to 2023 were obtained from the Hong Kong Observatory (HKO) station,¹⁹ chosen for its comprehensive data coverage. This dataset enabled us to identify and characterize heat wave events under the four different scenarios over the past decade.

Daily number of deaths

A key component of calculating excess heat-related deaths (EHD) is the deaths that occur under normal conditions (no heat wave). Expected daily mortality data, disaggregated by age-sex specific group, were sourced from the Hong Kong Life Tables covering the period from January 1, 2014, to December 31, 2023.^{20, 21} Subsequently, a time series of normal death counts was generated by multiplying the age-sex specific death rates by the corresponding population figures from the Hong Kong Census and Statistics Department.²²

Transformation of relative risks

We used estimates of the RRs due to heat from Liu et al. (2020)⁹ and Wang et al. (2019).¹¹ The theoretical basis for EHD involves a counterfactual comparison, contrasting observed conditions (e.g., heat exposure) with a hypothetical reference state (e.g., normal conditions without heat exposure). RR represents the ratio of the probability of death under heat exposure to that in a reference group under ordinary temperatures, as a key variable in calculating EHDs. All RRs are estimated at the level of the whole population, sex and age groups. Therefore, to make these RRs fit in the age-sex specific structure of Hong Kong Life Tables,^{20, 21} we adjusted them by combining the log-transformed relative risks for age and sex while accounting for the overall population risk. Mathematically, this is represented as **Equation 1**:

$$\ln(RR_{age,sex}) = \ln(RR_{age}) + \ln(RR_{sex}) - \ln(RR_{all}) \quad [1]$$

Subsequently, adjusted log-RRs are exponentiated, then reverted to the original RRs scale. Detailed adjusted RRs are provided in [Supplementary Table S1](#).

Adjusted RRs and their corresponding standard deviations are also logarithmically transformed. This transformation is essential as the RRs represent a ratio, and confidence intervals for their parameters are commonly estimated on the logarithmic scale, indicating a log-normal distribution.²⁴ Consequently, we generated 1000 values of log-normally distributed RRs, using a pseudo-random number generator in Stata by using **Equation 2** as follows:

$$RR = \exp((\text{invnorm}(\text{uniform}(\text{rand})) \times \log SD + \log RR)) \quad [2]$$

Estimation of excess mortality attributable to extreme heat

We subsequently calculated the EHDs for each single heat wave observed in the 10-year interval. The expected number of excess deaths attributable to extreme heat over various lag days for the t -th heat wave was determined using **Equation 3**:^{25, 26}

$$EHD_{ij} = (RR_{ijt} - 1) \times DD_{ij} \times L_t \quad [3]$$

Specifically, i denotes sex, j denotes age group. L_t signifies the duration in days of the t -th phase of a heat wave. RR_{ijt} is the relative risk of deaths for sex i , age group j , during the t -th phase of a heat wave, while DD_{ij} is the expected number of daily deaths in the counterfactual scenario of no heat wave. We generated a distribution of 1000 random estimates of the RRs using **Equation 2**, then drew from the distribution to estimate EHD_{ij} using Monte Carlo simulation. The simulated results are expressed as EHD_{ijn} , where n ranges from 1 to 1000, representing the Monte Carlo simulation index. This method is detailed in Zhao et al. (2024).²⁷

Finally, we multiplied the cumulative count of heat waves in Hong Kong for each year from 2014 to 2023 by EHD_{ij} to get estimates of annual excess heat-related deaths which could be added up to the total deaths by age and sex in a decade. We also reported the excess mortality as age-standardized excess death rates per 100,000 population. The age adjustment was based on the world standard population as specified in the revised age-standardization of rates by the World Health Organization - 2001 Edition.²⁸

The estimated excess deaths count and the age-standardized death rates across the last decade were accompanied by 95% confidence intervals (CIs). These CIs were obtained

using the normal approximation with the bootstrap method described by Pizzato et al. (2024),²⁹ employing 1000 samples generated through Monte Carlo simulation. Given that discounting can substantially affect mortality estimates, as seen in the Global Burden of Disease approach,³⁰ we present excess heat-related deaths and death rates without discounting or age weighting, treating all deaths equally irrespective of age or when they occur.

We used Stata SE (v.18.0) for all statistical analyses. All data used in this analysis were aggregated statistics at the population level. The study was not human subject's research.

Results

Summary statistics

During the study period in Hong Kong, average daily temperature was 24.12°C, ranging from 12°C to 31.1°C ([Supplementary Table S2](#)). As shown in Figure 1 different heat wave definitions differ in frequency and in total days of exposure over a decade. The HWD_Tcombined definition racked up the highest exposure at 365 days over 10 years, and the highest frequency of occurrences at 38 events over 10 years. The definition labeled HWD_Tavg occurred on 360 heat wave days but had the lowest frequency of 18 events. In contrast, HWD_Tmax recorded the fewest heat wave days at 183 and a frequency of 25 events. HWD_Tmin demonstrated a slightly higher frequency of 26 occurrences whereas it also had a limited duration of 190 heat wave days ([Figure 1](#)).

Excess Mortality in number and age-standardized death rate

Based on the definition of heat event, the estimates of total excess deaths attributable to heat waves ranged from 1,455 (95% CI: 1,098–1,812) to 3,238 (95% CI: 3,234–3,242) ([Table 2](#) and [Figure 2A](#)). Notably, the heat wave definition based on combined temperature threshold metrics (HWD_Tcombined) resulted in more than double the estimated excess deaths compared to estimates derived from mean daily temperature (HWD_Tavg) ([Table 2](#)). The HWD_Tavg definition gave the lowest cumulative estimates of excess heat-related deaths. See [Table 2](#) and [Figure 2A](#).

Figure 3 presents the annual trends in excess death rates (per 100,000 individuals) in Hong Kong from 2014 to 2023 under various heat wave definitions and compares these

rates to Hong Kong's age adjusted diabetes mortality. Under the HWD_Tmax and HWD_Tcombined definitions, the excess death rate attributable to heat waves exceeded that of diabetes in nearly half of the study period (from 2020 to 2023). Notably, under the HWD_Tcombined definition, the excess death rate was nearly twice that of diabetes in the recent three years (**Table 3**). Consequently, heat waves exhibiting high-risk patterns are poised to become the tenth leading cause of death in Hong Kong, even surpassing diabetes.³¹

Sub-group analysis

Our estimate of excess death counts by age and sex reflect the epidemiological variability from the original estimates of RRs (**Supplementary Table S3** and **Figure S4, S6**). Males experience nearly twice the number of excess deaths compared to females across most metrics, except for the nighttime heat definition (**Supplementary Figure S4A**). The magnitude of heat wave-related mortality risk increases with age, as individuals aged over 85 years appear to be the most vulnerable, with more than twice the number of deaths compared to other age groups across most metrics (**Supplementary Figure S4B**). Similar demographic variability is evident in the age-standardized excess death rates (**Supplementary Figure S5**).

Discussion

Overall, our estimates of excess deaths in Hong Kong under different heat wave metrics ranged from 1455 to 3238 over a 10-year period. This finding helps to compare the magnitude of risk established in previous studies^{10, 11, 12} to benchmark conditions like diabetes. Notably, the magnitude of excess heat deaths shows variability across different heat wave metrics, highlighting the complexity of this public health issue.

Furthermore, our investigation reveals that extreme heat, if considered a distinct condition, would emerge as a leading cause of mortality in Hong Kong, with death rates nearly comparable to those of diabetes over the past decade. Specifically, the excess death rate ranged from 2.95 to 5.09 per 100,000 in 2023, positioning it among the top 10 causes of death in the region.³¹

Importantly, our findings reaffirm previous research indicating that individuals aged over 64 years in Hong Kong exhibit the highest susceptibility to heat wave-related excess mortality. This vulnerability is evidenced by both the excess number of deaths

and the elevated death rates observed in this demographic. This well-known finding is attributed to a decreasing thermoregulatory capacity over time, as well as increased exposure to vasoactive medications for chronic diseases, which make the elderly more prone to health hazards during hot weather.^{32, 33}

Behavioral change could prevent excess heat deaths. Some elderly individuals in Hong Kong have shown reluctance to use effective cooling appliances to save on energy costs, even if air conditioners are installed in their homes.³⁴ Many flats still have poor ventilation design and small window sizes characteristic of residential buildings in Hong Kong.³⁵ Therefore, tailored protective measures based on population vulnerability and the specific characteristics of heat waves could prevent excess deaths. Clinicians including pharmacists, nurses, and physicians can be more proactive in assessing and counseling patients about their risk from age, co-morbidities, and medications. Social agencies conducting home visits can also engage in solution finding that is in harmony with cultural beliefs.

Strengths and Policy Implications

To the best of our knowledge, this study is the first to estimate excess heat-related mortality in Hong Kong, encompassing both the direct and indirect impacts of extreme heat. While previous research has primarily focused on the relationship between temperature and mortality through relative risk estimates, our study shifts the emphasis to the absolute number of excess deaths—a more tangible and actionable metrics for policymakers and the public. Using the best available local evidence, we generated these estimates while accounting for a range of scenarios to reflect the complex and multifaceted nature of heat exposure and its health impacts.

Under global warming, Hong Kong is projected to experience more hot days and fewer cold days in the future,³⁶ making it imperative for policymakers to develop adaptation strategies to protect vulnerable populations, especially in the context of the region's rapidly aging population. In response to such challenges, many countries have introduced Heat Action Plans (HAPs), which include early warning systems, public advisories, and emergency health measures to reduce heat-related morbidity and mortality.³⁷ Empirical evidence indicates that Heat Action Plans (HAPs) can effectively mitigate the impact of extreme temperatures. For instance, studies in Ahmedabad and

France reported significant reductions in all-cause mortality following HAP implementation, with Ahmedabad avoiding 2,380 deaths on heat alert days and France reporting 4,388 fewer deaths during the 2006 European Heatwave.^{38, 39} Similarly, a study in Montreal showed a reduction in daily mortality post-HAP, particularly among the elderly and lower socioeconomic status groups, thereby addressing heat-related health inequities.⁴⁰ Despite these successes, Hong Kong lacks a specific HAP tailored to its vulnerable population, highlighting the urgent need for an integrated heat-response strategy within the health and social care system.

Limitations

Using local evidence can be a double-edged sword. First, in the absence of a universally accepted standard for measuring human exposure to heat, our study provides four distinct estimates of excess heat-related deaths, each based on different definitions of heatwaves used in previous local studies.^{9, 11} These measures can vary significantly depending on how heatwaves are characterized, such as by average temperature,⁹ maximum temperature, or nighttime temperature, and how deaths occurring in the days or weeks following a heatwave are accounted for.¹¹ Consequently, there is no single, universally accepted method for attributing deaths to heat, resulting in a range of estimates rather than a definitive value. While it remains challenging to determine which estimates are the best, we offer these as a range, inviting readers to consider the broad spectrum of possible outcomes. Second, we assumed the entire territory of Hong Kong population is at equal risk of heat exposure, but heat levels vary spatially, and risk levels vary by education, occupation, socioeconomic status, and physical condition. Further stratification studies are needed to explore demographic heterogeneity.

Conclusion

This study estimated the excess death count attributable to heat in Hong Kong over the past decade and leverages the best available local evidence to provide a range of scenarios, each grounded in methodologies established by previous research. Notably, individuals aged over 65 years were disproportionately affected, underscoring the heightened vulnerability of the senior population to extreme heat.

Because heat deaths are theoretically preventable, our study highlights the critical

importance of developing territory-wide HAPs. Cities that convene multiple branches of government and civil society to tailor a local culturally appropriate approach have demonstrated success in controlling preventable deaths. Without such measures, extreme heat will continue to grow as a major cause of mortality in Hong Kong, posing significant public health challenges in the context of ongoing climate change.

Acknowledgments

Author Contributions

Concept or design: DMB

Acquisition of data: JL, CR

Analysis or interpretation of data: ZL

Drafting of the manuscript: ZL, YK

Critical revision of the manuscript for important intellectual content: CR, JL, DMB.

All data used in this paper is in the public domain and is listed in the citations.

CHATGPT-4o was used in the writing process to improve readability and language of the work, but not to analyze or interpret data or draw scientific conclusions.

Conflicts of interest

All authors have disclosed no conflicts of interest.

Declaration

The results of this study were presented in the Hong Kong College of Community Medicine Annual Scientific Meeting in Hong Kong (28 September 2024).

Ethics approval

This study did not require ethical approval as it exclusively used publicly available data obtained from previously published. No new data was collected from human participants, and all analyses were conducted on anonymized, aggregated datasets from public sources.

References

1. Allan, R.P., et al., *Intergovernmental Panel on Climate Change (IPCC). Summary for Policymakers*, in *Climate change 2021: The physical science basis. Contribution of working group I to the sixth assessment report of the intergovernmental panel on climate change*. 2023, Cambridge University Press. p. 3-32.
2. Hermanson, L., et al., *WMO global annual to decadal climate update: a prediction for 2021–25*. Bulletin of the American Meteorological Society, 2022. **103**(4): p. E1117-E1129.
3. Kovats, R.S. and S. Hajat, *Heat stress and public health: a critical review*. Annu. Rev. Public Health, 2008. **29**(1): p. 41-55.
4. Guo, Y., et al., *Heat wave and mortality: a multicountry, multicomunity study*. Environmental health perspectives, 2017. **125**(8): p. 087006.
5. Gasparrini, A. and B. Armstrong, *The impact of heat waves on mortality*. Epidemiology, 2011. **22**(1): p. 68-73.
6. Ho, J.Y., et al., *Urban heat island effect-related mortality under extreme heat and non-extreme heat scenarios: A 2010–2019 case study in Hong Kong*. Science of the Total Environment, 2023. **858**: p. 159791.
7. Zhou, B., D. Rybski, and J.P. Kropp, *The role of city size and urban form in the surface urban heat island*. Scientific reports, 2017. **7**(1): p. 4791.
8. Yi, W. and A.P. Chan, *Effects of temperature on mortality in Hong Kong: a time series analysis*. International journal of biometeorology, 2015. **59**: p. 927-936.
9. Liu, J., et al., *Cause-specific mortality attributable to cold and hot ambient temperatures in Hong Kong: a time-series study, 2006–2016*. Sustainable Cities and Society, 2020. **57**: p. 102131.
10. Leung, Y., K. Yip, and K. Yeung, *Relationship between thermal index and mortality in Hong Kong*. Meteorological Applications: A journal of forecasting, practical applications, training techniques and modelling, 2008. **15**(3): p. 399-409.
11. Wang, D., et al., *The impact of extremely hot weather events on all-cause mortality in a highly urbanized and densely populated subtropical city: A 10-year time-series study (2006–2015)*. Science of the Total Environment, 2019. **690**: p. 923-931.
12. Ho, H.C., et al., *Characterizing prolonged heat effects on mortality in a subtropical high-density city, Hong Kong*. International journal of biometeorology, 2017. **61**: p. 1935-1944.
13. Chan, E.Y.Y., et al., *A study of intracity variation of temperature-related mortality and socioeconomic status among the Chinese population in Hong Kong*. J Epidemiol Community Health, 2012. **66**(4): p. 322-327.
14. Weinberger, K.R., et al., *Estimating the number of excess deaths attributable to heat in 297 United States counties*. Environmental Epidemiology, 2020. **4**(3): p. e096.
15. Ho, J.Y., et al., *Compound hot-humid extreme events and their mortality associations during summer and shoulder months in a subtropical coastal city*. Sustainable Cities and Society, 2024: p. 106031.
16. Hajat, S. and T. Kosatky, *Heat-related mortality: a review and exploration of heterogeneity*. Journal of Epidemiology & Community Health, 2010. **64**(9): p. 753-760.
17. WHO, W.H.O., *The true death toll of COVID-19: Estimating global excess*

- mortality. World Health Organisation, 2021. **30**: p. 1.
18. Beaney, T., et al., *Excess mortality: the gold standard in measuring the impact of COVID-19 worldwide?* Journal of the Royal Society of Medicine, 2020. **113**(9): p. 329-334.
19. Observatory, H.K., *Daily Mean Temperature All Year 1886-2024*. 2024.
20. Census and Statistics Department, T.G.o.t.H.K.S., *Hong Kong Life Tables 2014–2069*. 2020.
21. Census and Statistics Department, T.G.o.t.H.K.S., *Hong Kong Life Tables 2016–2046*. 2023.
22. Department, H.K.C.a.S., *Table 110-01002: Population by Sex and Age Group (Released 20 Feb 2024)*. 2024.
23. Mock, C., S. Pilcher, and R. Maier, *Comparison of the costs of acute treatment for gunshot and stab wounds: further evidence of the need for firearms control*. Journal of Trauma and Acute Care Surgery, 1994. **36**(4): p. 516-522.
24. Briggs, A., M. Sculpher, and K. Claxton, *Decision modelling for health economic evaluation*. 2006: Oup Oxford.
25. Peng, R.D., et al., *Toward a quantitative estimate of future heat wave mortality under global climate change*. Environmental health perspectives, 2011. **119**(5): p. 701-706.
26. Wu, J., et al., *Estimation and uncertainty analysis of impacts of future heat waves on mortality in the eastern United States*. Environmental health perspectives, 2014. **122**(1): p. 10-16.
27. Zhao, Q., et al., *Global, regional, and national burden of heatwave-related mortality from 1990 to 2019: A three-stage modelling study*. PLoS medicine, 2024. **21**(5): p. e1004364.
28. Ahmad, O.B., et al., *Age standardization of rates: a new WHO standard*. Geneva: World Health Organization, 2001. **9**(10): p. 1-14.
29. Pizzato, M., et al., *Impact of COVID-19 on total excess mortality and geographic disparities in Europe, 2020–2023: a spatio-temporal analysis*. The Lancet Regional Health–Europe, 2024.
30. Murray, C., *A Comprehensive Assessment of Mortality and Disability from Disease, Injuries and Risk factors in 1990 and projected to 2020*. The global burden of disease, 1994. **1**.
31. Centre for Health Protection, D.o.H., The Government of the Hong Kong Special Administrative Region., *Age-standardised Death Rates by Leading Causes of Death, 2001-2023*. 2024.
32. Åström, D.O., F. Bertil, and R. Joacim, *Heat wave impact on morbidity and mortality in the elderly population: a review of recent studies*. Maturitas, 2011. **69**(2): p. 99-105.
33. Foroni, M., et al., *A retrospective study on heat-related mortality in an elderly population during the 2003 heat wave in Modena, Italy: the Argento Project*. The Journals of Gerontology Series A: Biological Sciences and Medical Sciences, 2007. **62**(6): p. 647-651.
34. Yip, A.O.-n., D.N.-y. Mah, and L.B. Barber, *Revealing hidden energy poverty in Hong Kong: a multi-dimensional framework for examining and understanding energy poverty*. Local Environment, 2020. **25**(7): p. 473-491.
35. Mah, J.C., et al., *Social factors influencing utilization of home care in community-dwelling older adults: a scoping review*. BMC geriatrics, 2021. **21**: p. 1-21.
36. Leung, Y., *Climate Change in Hong Kong*. 2004: Hong Kong Observatory.

37. Matthies, F., et al., *Heat-health action plans: guidance*. 2008: World Health Organization.
38. Hess, J.J., et al., *Building Resilience to Climate Change: Pilot Evaluation of the Impact of India's First Heat Action Plan on All-Cause Mortality*. *Journal of environmental and public health*, 2018. **2018**(1): p. 7973519.
39. Fouillet, A., et al., *Has the impact of heat waves on mortality changed in France since the European heat wave of summer 2003? A study of the 2006 heat wave*. *International journal of epidemiology*, 2008. **37**(2): p. 309-317.
40. Benmarhnia, T., et al., *A difference-in-differences approach to assess the effect of a heat action plan on heat-related mortality, and differences in effectiveness according to sex, age, and socioeconomic status (Montreal, Quebec)*. *Environmental health perspectives*, 2016. **124**(11): p. 1694-1699.

Tables

Table 1. Heat wave definitions used as the basis for the present analysis.

Type	Name	Heat Wave Criteria	Estimation of RRs	Source
HWD_Tavg	Daily average temperature-based definition	• A consecutive series of 20 days beginning with a day that daily mean temperature $\geq 30.60^{\circ}\text{C}$; • An independent heat wave is classified when the first day is at least 20 days apart from the last day exceeding the threshold.	• RRs were calculated on a daily basis. • For each independent heat wave, RRs were assessed across multiple lag periods, divided into three distinct phases: - 0th day: Captures the immediate effects of the heat wave. - 1st to 5th days: Addresses delayed effects following the heat wave. - 6th to 21st days: Evaluates long-term effects associated with the heat wave.	Liu et al. (2020). ⁹
HWD_Tmax	Daily maximum temperature-based definition	• A single day when the daily maximum temperature $\geq 33^{\circ}\text{C}$ is considered as a very hot day. • A consecutive series of VHDs with lengths longer than 5 days (5HD).	• Relative risks (RRs) were calculated on a daily basis. • No detailed statistics are provided on the lagged effects of heat waves following the 5HD pattern.	Wang et al. (2019). ¹¹
HWD_Tmin	Daily minimum temperature-based definition	• A single day when the daily minimum temperature $\geq 28^{\circ}\text{C}$ is considered as a very hot	• Relative risks (RRs) were calculated on a daily basis.	Wang et al. (2019). ¹¹

HWD_Tcombined	Combined daily temperature threshold-based definition	night. • A consecutive series of VHNs with lengths longer than 5 days (5HN). • Defined as two consecutive days with maximum temperatures of 33°C or higher and three consecutive nights with minimum temperatures of 28°C or higher (2D3N). • The mortality effects from 2D3N heat waves show a significant five-day lag after their onset..	• No detailed statistics are provided on the lagged effects of heat waves following the 5HD pattern. • Relative risks (RRs) were calculated on a daily basis. • For each independent heat wave, RRs were assessed across multiple lag periods, divided into two distinct phases: - Days observed as part of 2D3N event - Subsequent five days following this event	Wang et al. (2019). ¹¹
---------------	---	---	--	-----------------------------------

Table 2. Cumulative excess deaths attributable to heat waves over the 10-year period under different heat wave definitions.

Year	HWD_Tavg	HWD_Tmax	HWD_Tmin	HWD_Tcombined
Total	1455 (1098, 1812)	2262 (2170, 2353)	1553 (1549, 1557)	3238 (3234, 3242)
2014	82 (64, 100)	34 (32, 35)	43 (41, 45)	128 (125, 131)
2015	160 (125, 196)	97 (93, 101)	125 (119, 131)	150 (147, 154)
2016	158 (121, 195)	108 (104, 113)	100 (95, 105)	240 (234, 245)
2017	75 (57, 94)	43 (41, 44)	168 (159, 176)	280 (273, 286)
2018	147 (109, 185)	367 (351, 382)	0 (0, 0)	130 (127, 133)
2019	75 (56, 94)	103 (99, 107)	190 (180, 199)	314 (306, 321)
2020	149 (110, 188)	364 (349, 379)	278 (265, 292)	337 (328, 345)
2021	148 (111, 186)	368 (353, 384)	176 (167, 184)	538 (525, 551)
2022	202 (155, 249)	496 (477, 515)	261 (248, 274)	547 (533, 560)
2023	257 (193, 322)	282 (271, 293)	213 (203, 224)	575 (562, 589)

Table 3. Annual Age-standardized excess death Rate per 100,000 population in the years 2014-2023 period

Year	HWD_Tavg	HWD_Tmax	HWD_Tmin	HWD_Tcombined	Diabetes
2014	1.18 (0.98, 1.38)	0.45 (0.44, 0.47)	0.56 (0.55, 0.56)	1.60 (1.59, 1.61)	2.5
2015	2.30 (1.89, 2.71)	1.24 (1.2, 1.29)	1.53 (1.52, 1.54)	1.80 (1.80, 1.81)	3.2
2016	2.25 (1.85, 2.65)	1.34 (1.28, 1.39)	1.19 (1.18, 1.20)	2.76 (2.75, 2.77)	3.1
2017	0.88 (1.74, 0.87)	0.44 (0.42, 0.46)	2.06 (2.05, 2.08)	2.98 (2.97, 2.99)	2.3
2018	1.74 (1.35, 2.13)	3.77 (3.61, 3.94)	0.00 (0,0)	1.31 (1.31, 1.32)	2.8
2019	0.87 (0.69, 1.06)	1.02 (0.98, 1.07)	2.15 (2.13, 2.17)	3.07 (3.05, 3.08)	2.8
2020	1.78 (1.39, 2.16)	3.56 (3.41, 3.72)	3.12 (3.09, 3.14)	3.26 (3.24, 3.27)	3.3
2021	2.00 (1.65, 2.35)	3.53 (3.39, 3.68)	1.80 (1.79, 1.82)	5.23 (5.21, 5.25)	2.9
2022	2.46 (2.04, 2.87)	4.54 (4.35, 4.74)	2.62 (2.6, 2.64)	5.07 (5.05, 5.09)	3.5
2023	2.95 (2.41, 3.5)	2.63 (2.52, 2.75)	2.05 (2.04, 2.07)	5.09 (5.07, 5.12)	2.8

Supplementary Materials

Supplementary Table S1. Age-sex specific Relative risk of extreme hot temperature on mortality and sub-categories over multiple lag days in Hong Kong: under four different heat wave definitions.

		HWD_Tavg	
Lag days	0	1-5	6-21
Female			
0-59	1.010 (0.962, 1.06)	0.998 (0.916, 1.087)	0.976 (0.827, 1.15)
60-74	0.993 (0.843, 1.046)	1.011 (0.932, 1.097)	1.057 (0.909, 1.228)
74-85	1.019 (0.978, 1.062)	1.023 (0.952, 1.099)	1.146 (0.975, 1.347)
85+	1.022 (0.961, 1.088)	0.976 (0.85, 1.122)	1.020 (0.838, 1.241)
Male			
0-59	1.036 (0.976, 1.099)	1.014 (0.922, 1.115)	0.929 (0.779, 1.108)
60-74	1.019 (0.957, 1.084)	1.027 (0.937, 1.126)	1.006 (0.855, 1.185)
74-85	1.045 (0.991, 1.103)	1.039 (0.957, 1.129)	1.091 (0.917, 1.298)
85+	1.048 (0.977, 1.126)	0.992 (0.858, 1.146)	0.972 (0.791, 1.194)
		HWD_Tmax	
Male			

0-19	1.063 (0.835, 1.353)
20-64	1.032 (0.938, 1.135)
65+	1.055 (0.983 .1.32)

Female

0-19	1.036 (0.812, 1.323)
20-64	1.005 (0.908, 1.114)
65+	1.028 (0.949, 1.113)

HWD_Tmin

Male

0-19	1.24 (0.99, 1.55)
20-64	1.05 (0.99, 1.11)
65+	1.03 (0.97, 1.10)

Female

0-19	1.26 (1.00, 1.57)
20-64	1.06 (0.99, 1.13)
65+	1.05 (0.97, 1.13)

HWD_Tcombined

Lag days

0

1-5

Male		1.038 (1.014, 1.063)
0-19	0.972 (0.790, 1.195)	-
20-64	1.016 (0.926, 1.115)	-
65+	1.038 (0.968, 1.113)	-
Female		1.038 (1.014, 1.063)
0-19	1.012 (0.821, 1.248)	-
20-64	1.059 (0.961, 1.167)	-
65+	1.081 (1.003, 1.166)	-

Supplementary Table S2. Descriptive statistics of daily deaths, mean temperature

Variables	Mean (SD)	Minimum	25th percentile	Median	75th percentile	Maximum
Daily death counts						
Total	183.8206	164.4876	174.1308	181.1151	182.5402	224.0055
<u>Gender</u>	(16.97849)					
Female	90.44614 (7.739586)	80.48383	84.85381	89.64177	93.87144	106.095
Male	93.37445 (9.898557)	84.00381	87.14879	89.80891	93.4278	117.9105
<u>Age</u>						
0-64	25.72308 (0.7905529)	24.83212	25.3142	25.49419	26.04878	27.72061
65-74	23.46998 (4.01261)	25.74804	29.06804	32.11756	34.30605	42.13499
75-84	34.93737 (2.832619)	31.96074	32.67887	34.33803	36.31163	41.22369
85+	99.69016 (11.82514)	83.57468	93.54366	98.50468	101.0715	123.5365
Mean Temperature	24.12221 (5.09736)	12	20	25.1	28.8	31.1

Supplementary Table S3. Demographic heterogeneity in total number of excess heat-related deaths: by age and sex groups

	HWD_Tavg	HWD_Tmax	HWD_Tmin	HWD_Tcombined
Sex				
Female	-11 (-201, 178)	802 (755, 849)	877 (837, 918)	1154 (1117, 1192)
Male	1466 (1294, 1638)	1459 (1413, 1505)	676 (639, 713)	2083 (2042, 2124)
Age				
0-64	-45 (-88, -3)	241 (224, 258)	281 (270, 291)	382 (368, 397)
65-74	320 (281, 360)	342 (331, 353)	183 (172, 193)	481 (471, 491)
75-84	1331 (1269, 1393)	465 (449, 480)	263 (249, 276)	656 (643, 669)
85+	-152 (-366, 63)	1214 (1167, 1260)	828 (784, 871)	1719 (1679, 1759)

Figure 1. The descriptive statistics of heatwaves. A: Annual sum of frequency of heatwaves in Hong Kong, during 2014–2023; B: Annual counts of heat wave days in Hong Kong, during 2014–2023.

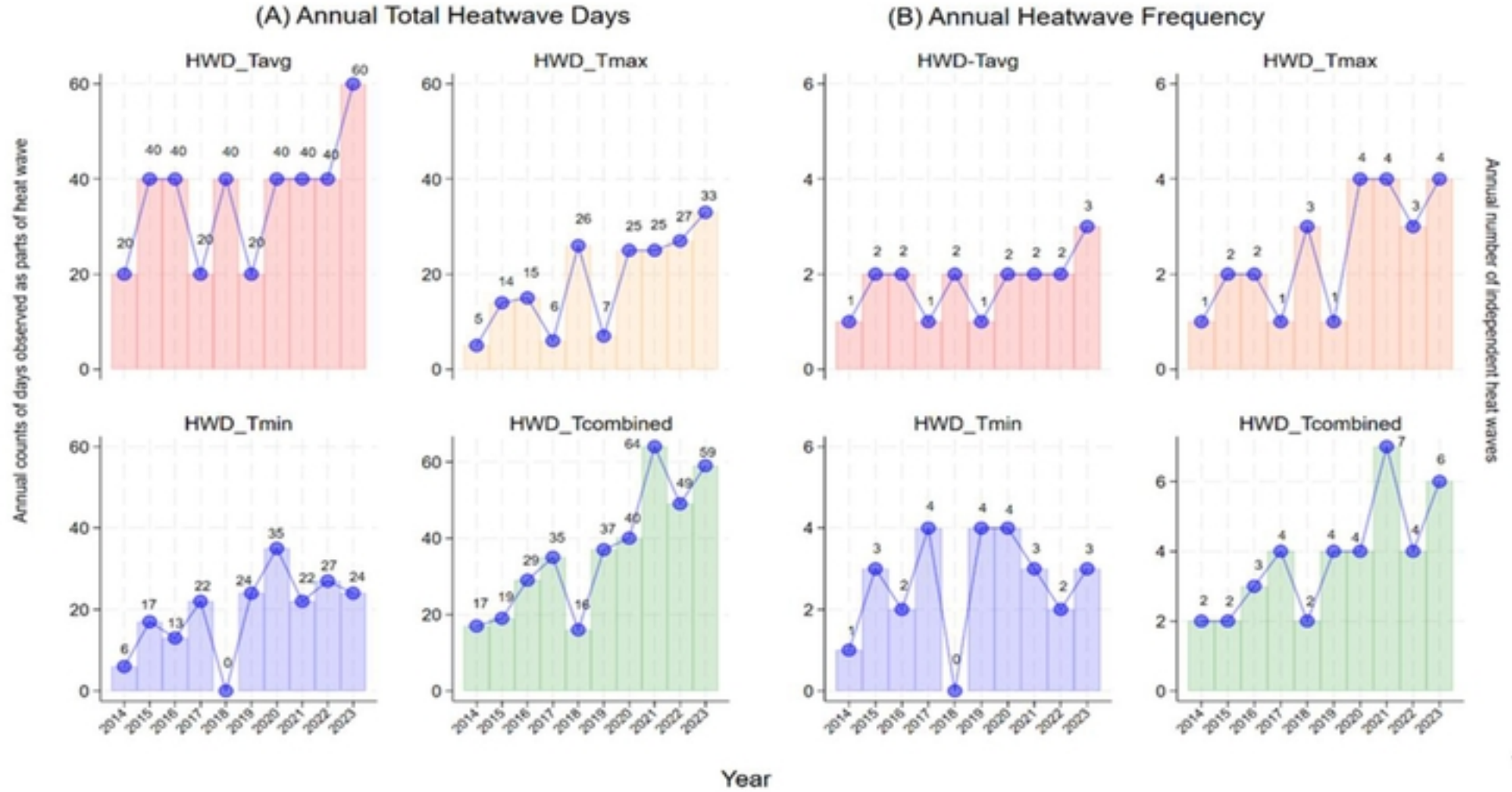
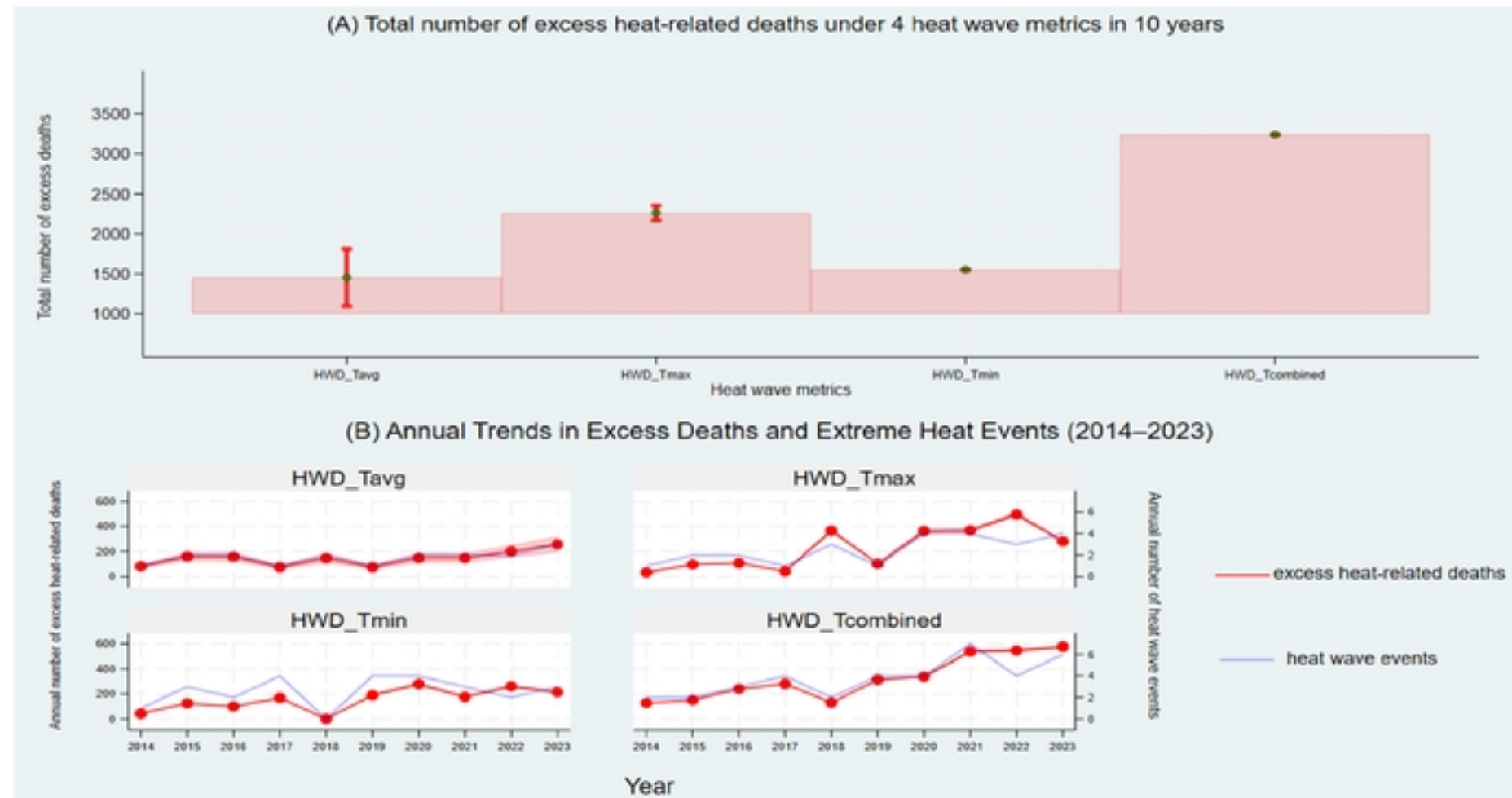
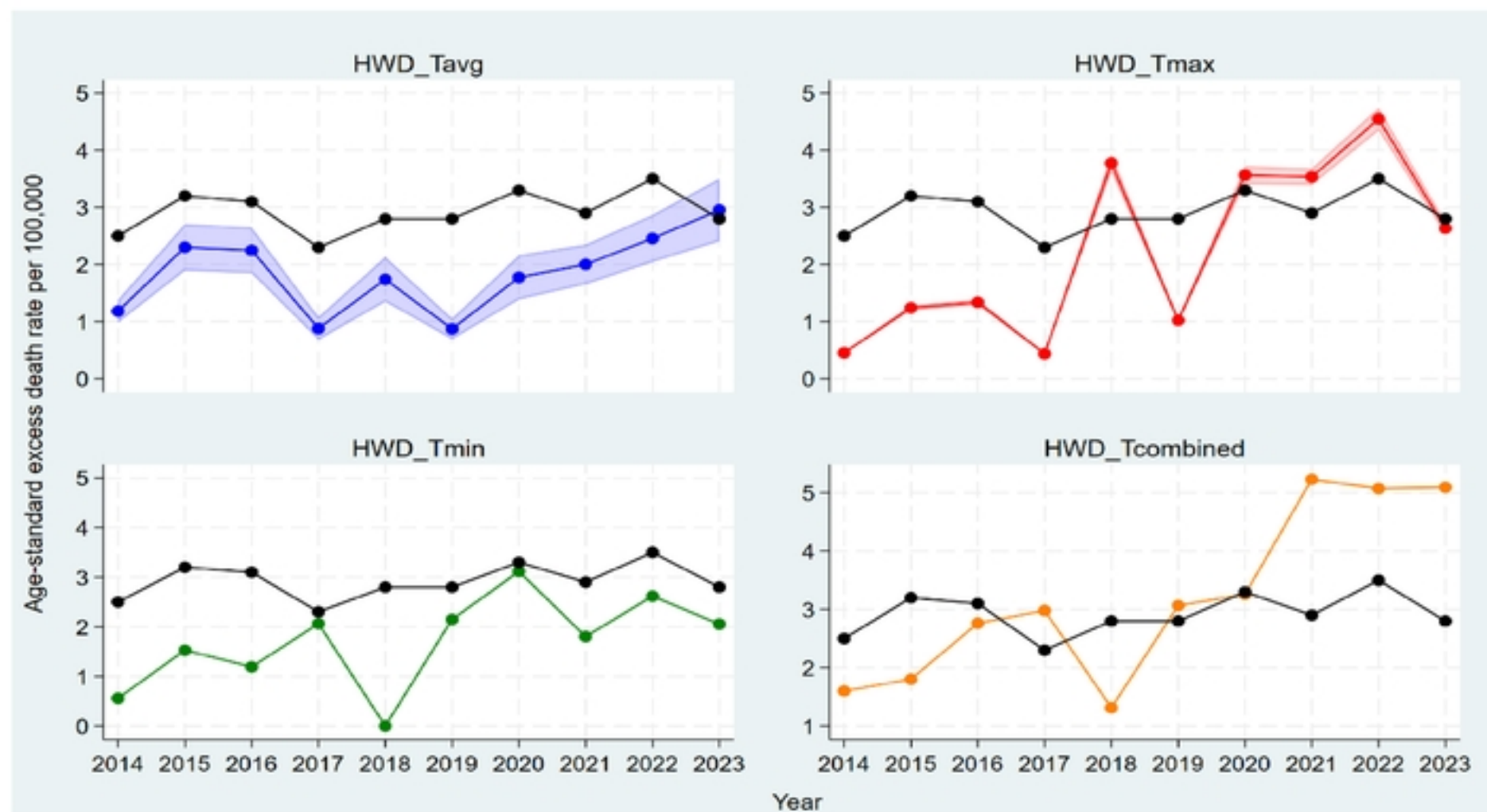


Figure 2. Excess heat-related deaths (2014–2023) under 4 definitions. A: Total number of excess heat-related deaths across 10 years; B: Annual number of excess heat-related deaths



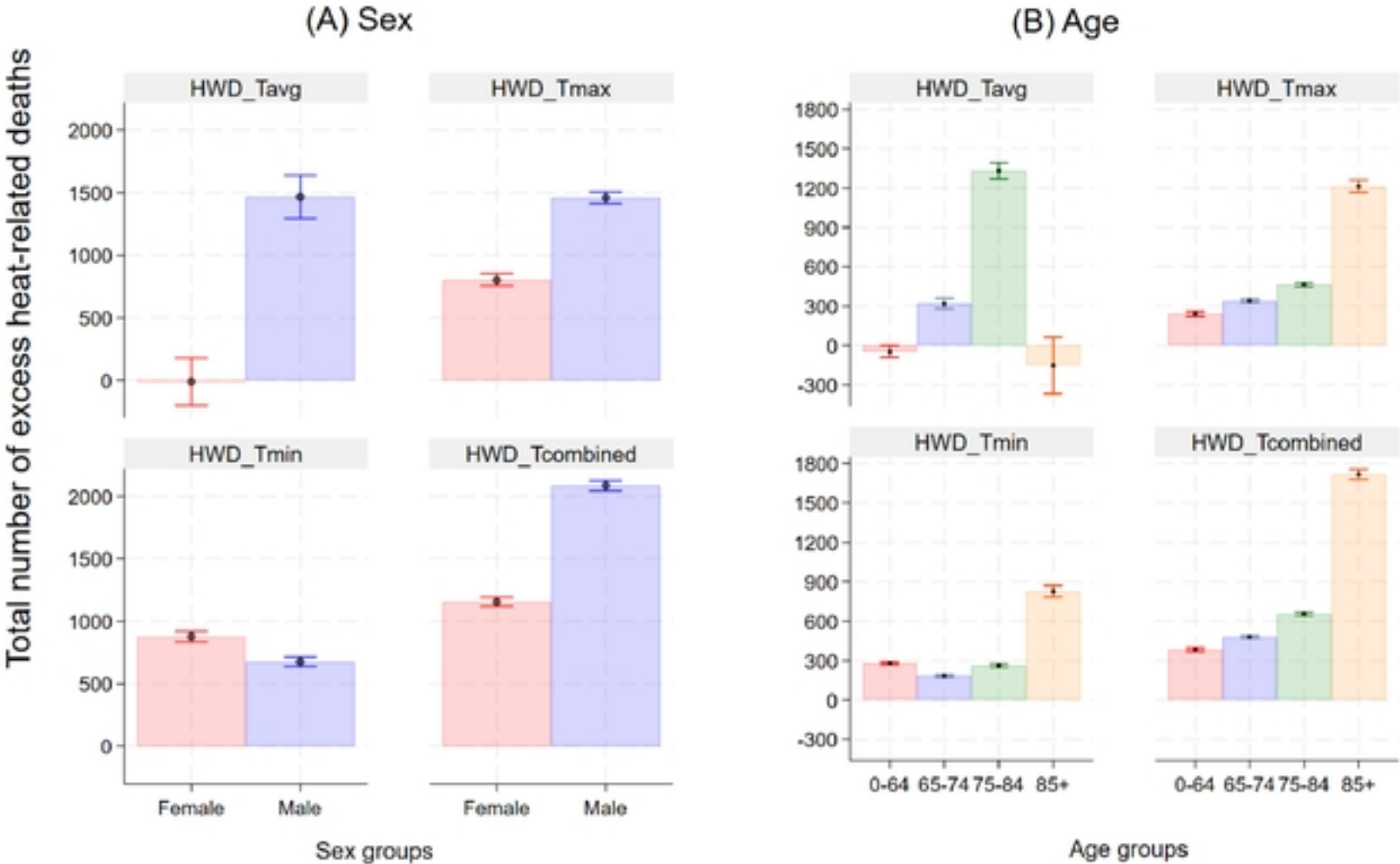
Notes: In Fig2A, the vertical lines extending from the top of each bar indicate the 95% CIs derived from Monte Carlo simulation. In Fig2B, the shaded areas around the lines indicate the 95% CIs derived from Monte Carlo simulation, providing a measure of the statistical uncertainty around the annual excess death estimates.

Figure 3. Annual Trends in Age-Standardized Excess Death Rates in Hong Kong (2014-2023)



Note: The black line depicts the age-standardized mortality rate for diabetes in Hong Kong, serving as a comparative benchmark for heat wave-related excess deaths.

Supplementary Figure S4. Demographic heterogeneity in total number of excess heat-related deaths: by age and sex groups



Supplementary Figure S5. Demographic Heterogeneity in age-standardized mortality rates: by age and sex groups

